Verification of Neural Networks as a Replacement for LUTs in ACAS-Xu

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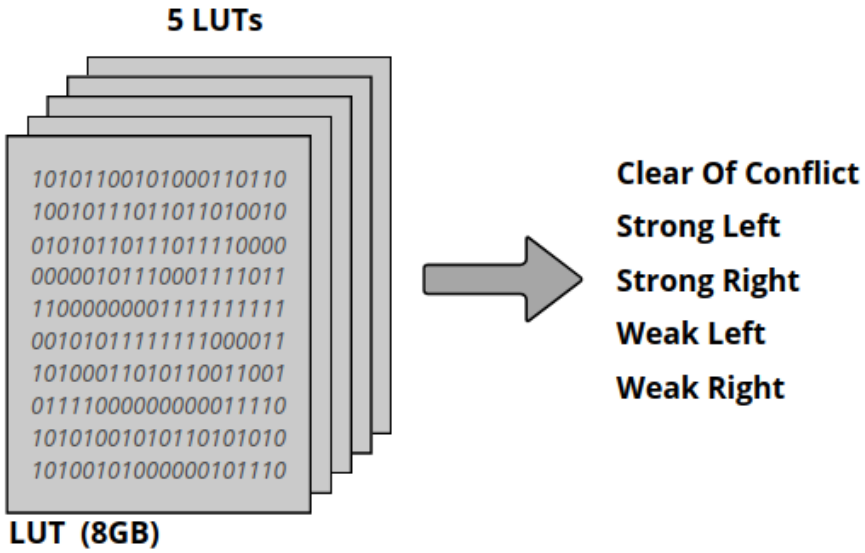
Context & Introduction

Anti-collision systems for air traffic are complex but essential

- TCAS since the 80s.

- ACAS-X new generation :
  - ACAS-Xa
  - ACAS-Xo
  - **ACAS-Xu**
  - …

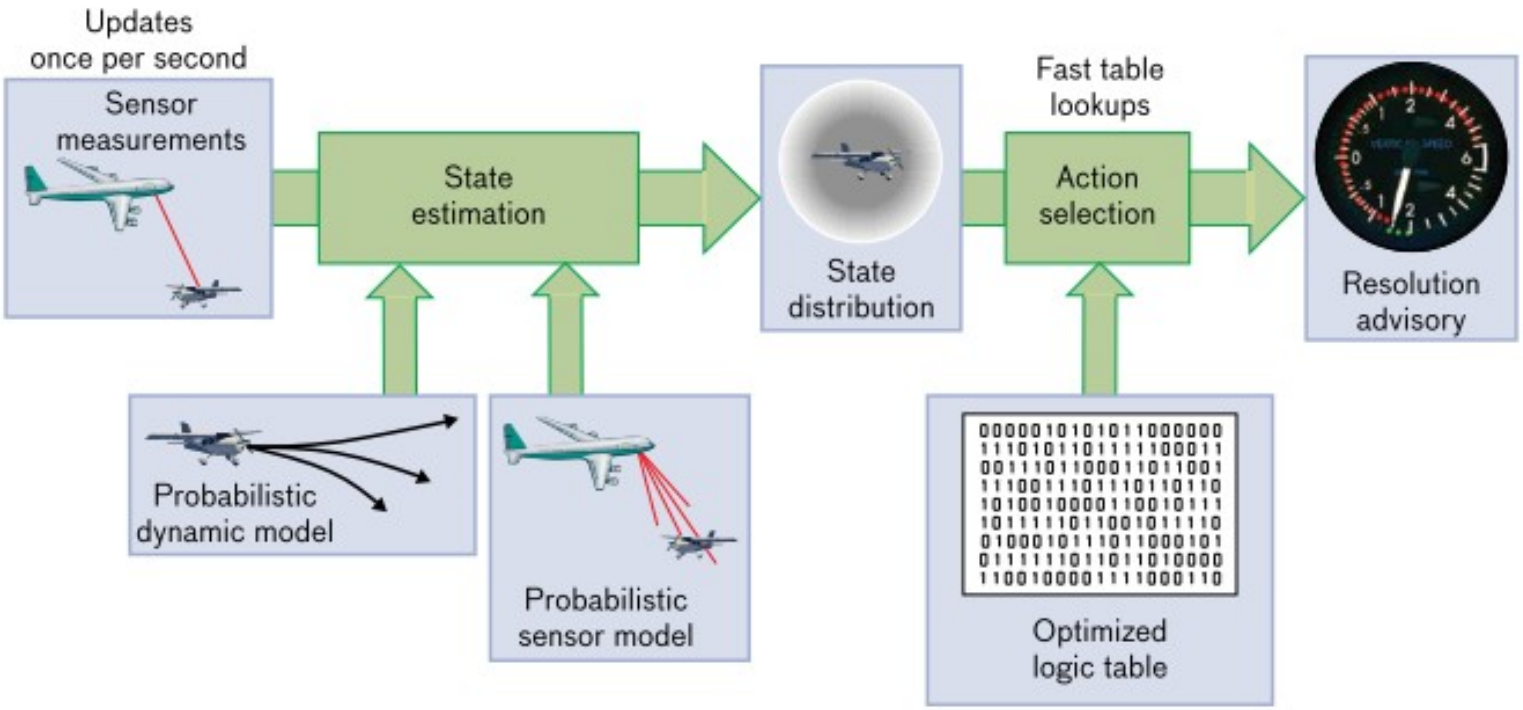
Context & Introduction



These systems are using **Look Up Tables** (LUT) for action selection.

One emerging idea was to update the technology : From LUT to **Neural Network.**

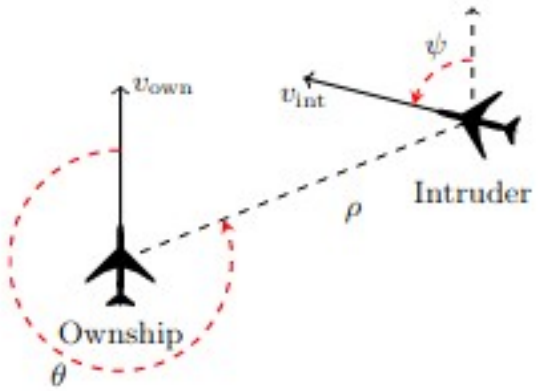
ACAS Presentation



The Look Up Tables

Big array of 11 columns (& ~95M lines).

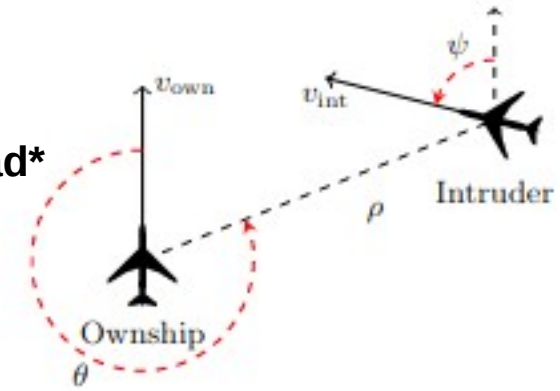
- 6 for the flight and geometric variables.
- 5 for the costs of each possible action (COC, SL, SR, WL, WR)



τ	θ	ψ	V_own	V_int	ρ	Cost_COC	Cost_SL	Cost_SR	Cost_WL	Cost_WR
0.0	-3.14	-3.14	50.0	100	499	24017	24017	24081	24001	24065
32.2	-3.14	0.0	50.0	142	3024	22913	22912	23040	22881	23008
41.7	0.0	3.14	87.6	807	45202	22645	22644	22772	22613	22741
101	3.14	1.5	87.6	1069	763118	20844	20832	21340	20700	21208

The Action Selection

- **Vertical Action :** Multi-Linear Interpolation & Minimal Cost
- **Horizontal Action :** Nearest Neighbor approach & **Lookahead***



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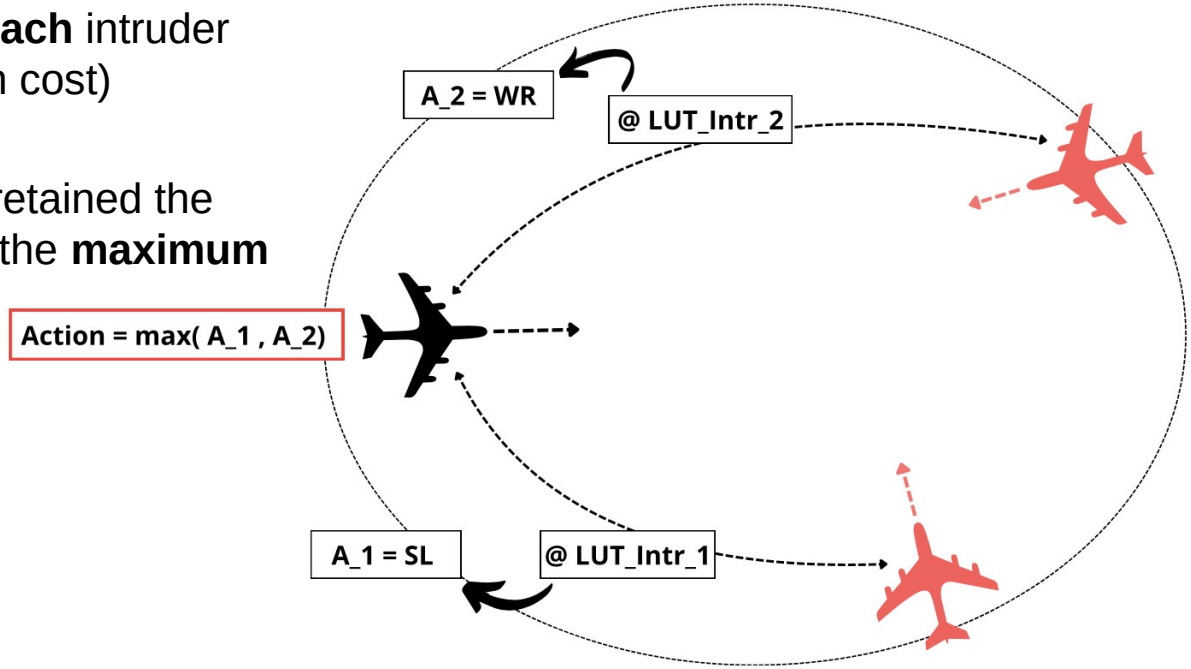
* : Complex mechanism whithout enouh info, futur work ?

The Action Selection : Multi-Intruder

- Call the tables to determine the avoidance **action** to be performed for **each** intruder (the action with the minimum cost)

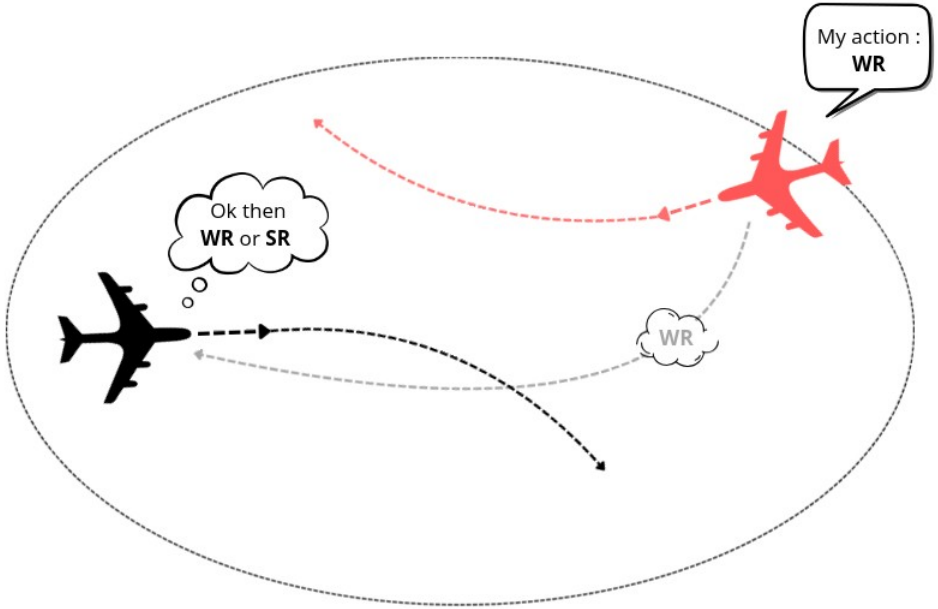
- Then among all the actions retained the system selects the one with the **maximum** cost.



The Action Selection : Cooperative-flight

The aircraft communicate with each other and exchange the actions taken.

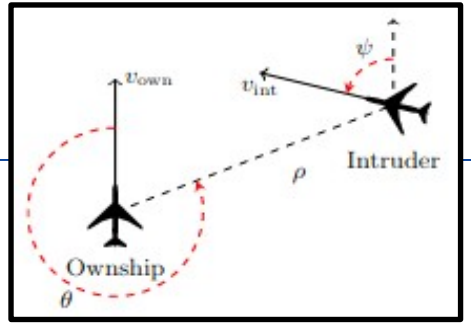
- Before executing the chosen action, the system first checks if a message has been received ; the aircraft coordinate their actions.



Approximations and Hypotheses

- We consider **Horizontal plan** problems only.
- We consider the ownship and the intruders in a **non-cooperative** flight. There is no communication about their actions.
- We do not consider the **“lookahead”** mechanism for now.
- Let’s focus on **single-intruder** scenarios.

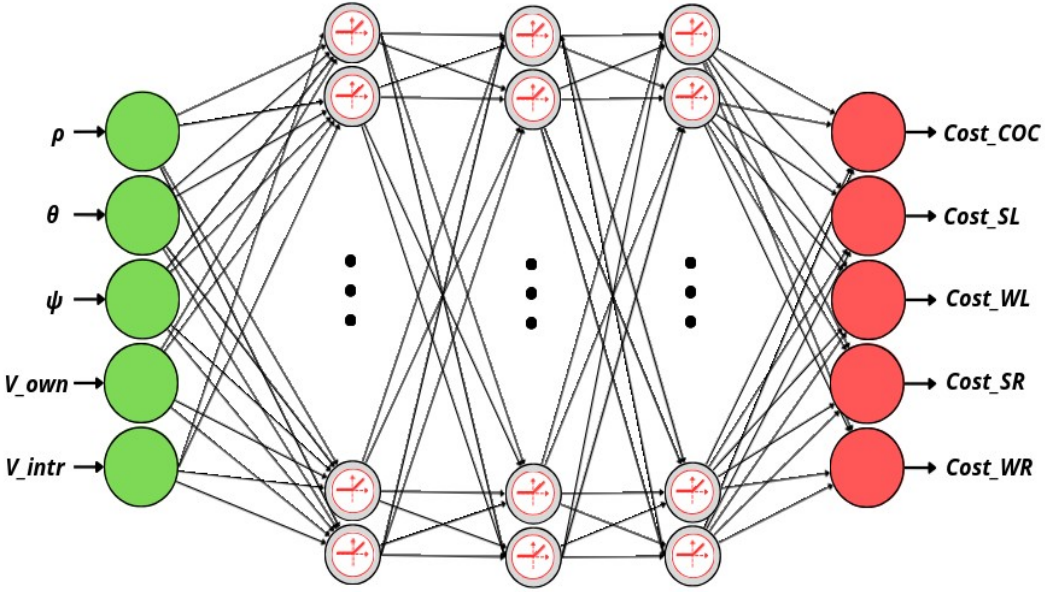
ACAS-Xu compression in Neural Network



- Fully connected architecture, with ReLU activation functions.
- 5 inputs : 5 states parameters (discretization on τ).
- 5 outputs : costs of the next actions.

Verification :

Ensuring that the NN meet specified properties and behave correctly under all possible inputs.



Reference : Julian, K. D., Kochenderfer, M. J., & Owen, M. P. (2018). Deep Neural Network Compression for Aircraft Collision Avoidance Systems

ACAS-Xu compression in Neural Network

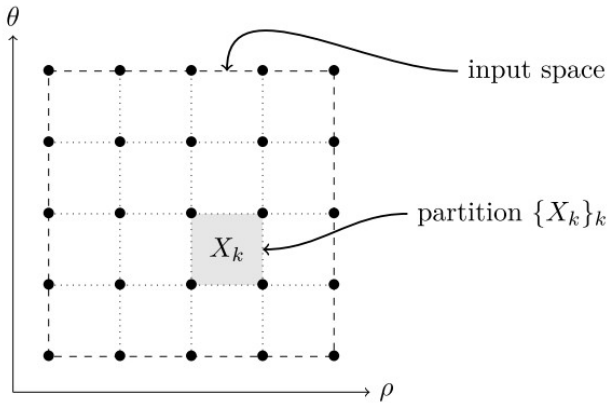
	TOPOLOGY	NUMBER OF NETWORKS	DETAILS
JULIAN NETWORKS	(5, 50, 50, 50, 50, 50, 50, 5)	45 NN : discretization on τ and a_prev	The first who introduced the neural network policy for the ACAS Xu
ADVERSERIAL NETWORKS	(5, 100, 100, 100, 5)	5 NN : discretization on a_prev ; τ = 0	Trained with the aim of satisfying the property defined in Arthur Clavière Phd *
CORNERS NETWORKS	(5, 100, 100, 100, 5)	5 NN : discretization on a_prev ; τ = 0	Trained with the aim of satisfying the property defined in Arthur Clavière Phd *
1-LIPSCHITZ NETWORKS	(5, 50, 50, 50, 50, 50, 50, 5)	5 NN : discretization on a_prev ; τ = 0	Trained with the aim of imposing a Lipschitz constant equal to 1

* : Clavière, A., & Benedikt, M. (2023). Safety verification of neural network based systems using formal methods.

Neural Network Verification for ACAS-Xu

Approach A : Verification of the compliance with the LUT Points

- Checking the NN results directly on the points from the LUT.
- Checking using an approximation property around the LUT points (Arthur Clavière thesis *).



Approach B : Verification of the good behavior through Scenarios

- Checking the consistency of the NN response across various scenarios.
- Using domain-specific properties to ensure consistency and reliability. Taken from the original ** paper and extended.

* : Clavière, A., & Benedikt, M. (2023). Safety verification of neural network based systems using formal methods.
** : Katz, G., Barrett, C., Dill, D., Julian, K., & Kochenderfer, M. (2017). Reluplex: An Efficient SMT Solver for Verifying Deep Neural Networks. http://arxiv.org/abs/1702.01135

ACAS-Xu Verification : Approach A

Proportion of COC actions in the LUTs : **~ 90%**

➔ Verification of the conformity of the NN on the LUT Points which are not COC.

	JULIAN NETWORKS	ADVERSERIAL NETWORKS	CORNERS NETWORKS	1-LIPSCHITZ NETWORKS
LUT (A_PREV = COC)	57.43 %	50.78 %	84.78 %	0.0 %
LUT (A_PREV = SR)	59.57 %	51.61 %	88.703 %	0.01 %
LUT (A_PREV = WL)	68.53 %	56.35 %	87.74 %	0.08 %

ACAS-Xu Verification : Approach B

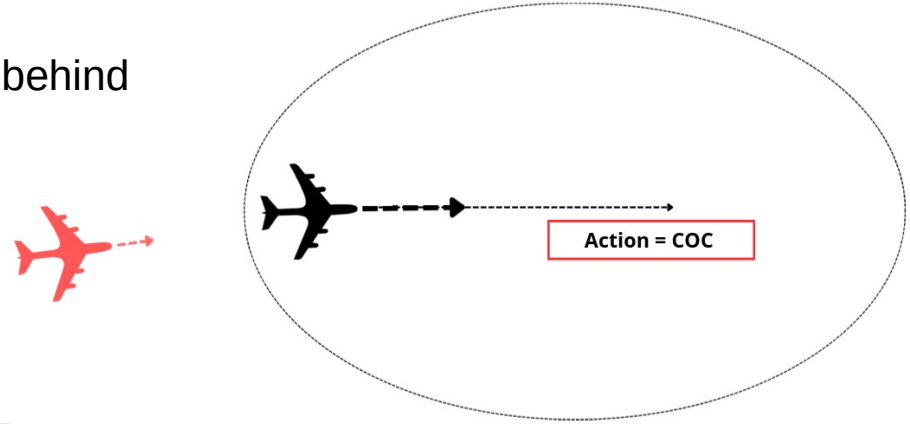
Definition of domain-specific properties which assess NN behavior in operational scenarios.

We consider Single-Intruder Avoidance Scenarios only :

- **Group 1 :** Scenarios where no avoidance maneuver is required. The system should recognize the absence of a collision threat and correctly return COC.

  Example :

  If the intruder is approaching the ownship from behind at a lower speed, the network will advise COC.



ACAS-Xu Verification : Approach B

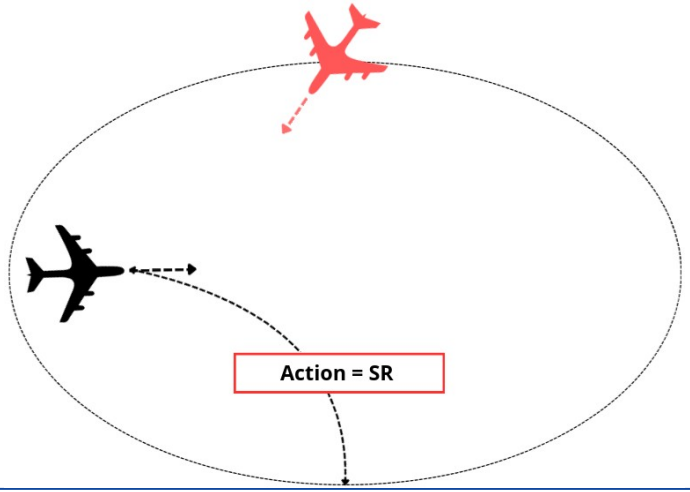
Definition of domain-specific properties which assess NN behavior in operational scenarios.

We consider Single-Intruder Avoidance Scenarios only :

- **Group 2 :** Scenarios where the system must actively recommend an avoidance maneuver. Returning COC would indicate a failure to respond appropriately to a potential collision risk.

  Example :

  If the intruder is near and approaching from the left, the network advises "Strong Right" (SR).



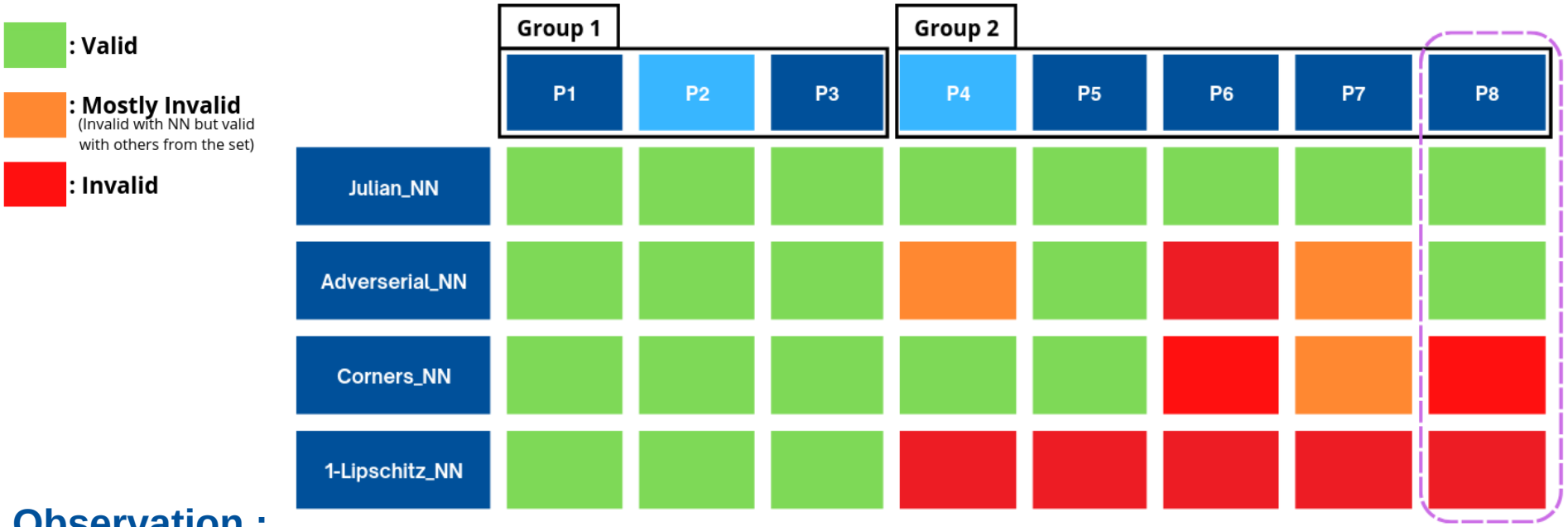
ACAS-Xu Verification : Approach B

What we are using :

- **Marabou** through **CAISAR**

Marabou : Formal verification tool for DNN, combining SMT, MILP, and the Reluplex algorithm to ensure robustness and safety in critical applications.

ACAS-Xu Verification : Approach B



Observation :

- **Group 1 :** 100 % Valid. COC as a response is always verified.
- **Group 2 :** Highlights the complexity to give another advisory than COC.

Conclusion

We have highlighted the difficulty of approximating the ACAS-Xu.

The main problem lies in the **poor distribution of actions** in the LUT. Neural networks tend to struggle in situations where avoidance actions are expected.

A good performance when evaluating specific points in the LUT (**Approach A**) does not guarantee a good approximation of the system's functioning (**Approach B**).

➔ Importance of building a **spec with domain-specific properties**.

Perspectives :

Pushing forward and proposing new high-level properties that take into account other aspects of the system (**multi-intruder** or **cooperative** scenarios.)

In the end, are **NN** truly the most relevant tools for approximating the entire system ?

Thank you for listening!
Questions ?

Neural Network Verification for ACAS-Xu

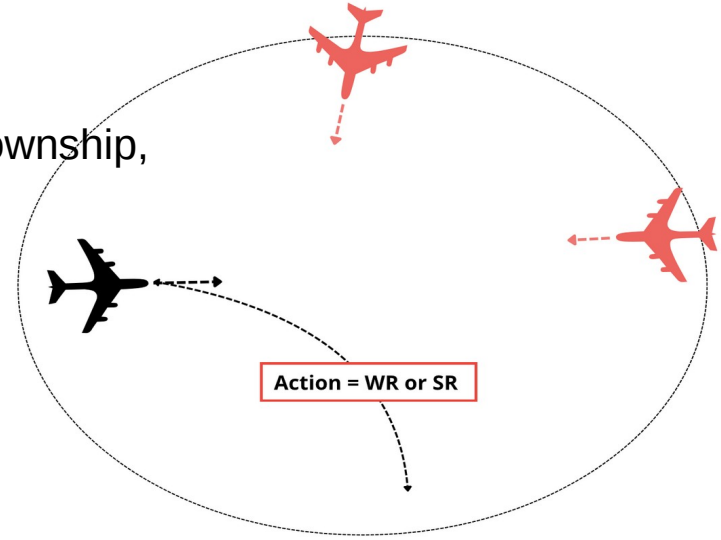
What we are doing :

- Specification of multi-intruder properties.
- Verification of these properties on CAISAR (Marabou / Nnenum / α-β-CROWN).

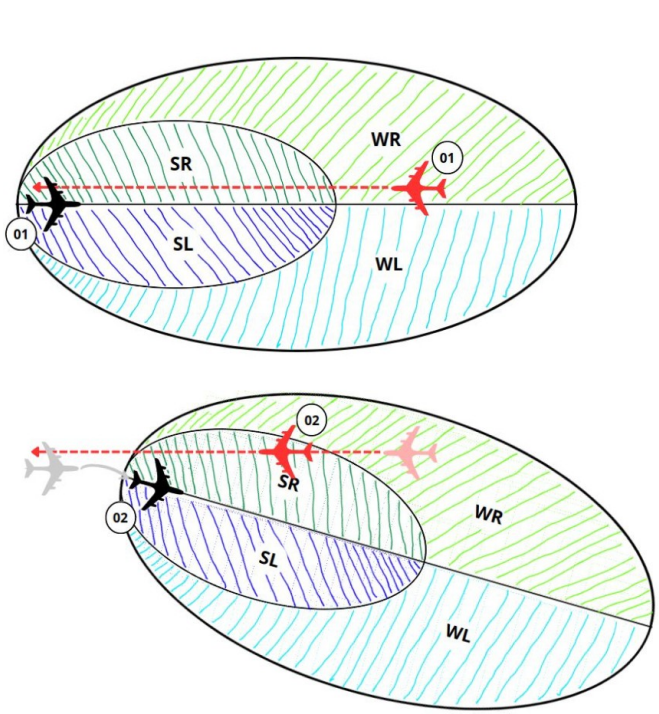
Example of multi-intruder properties :

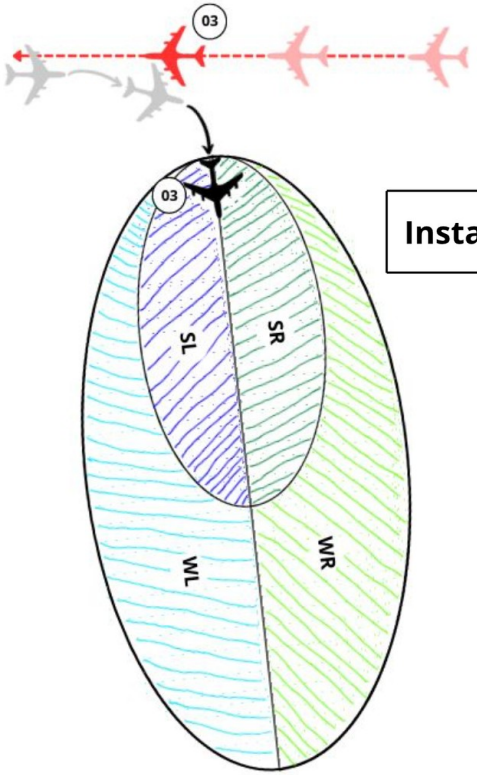
Description :

If intruder_1 is directly ahead and moving towards the ownship,
and the intruder_2 is approaching from the left,
the network advises a right turn (“WR” or “SR”).



Anexe - The “Lookahead” mechanism





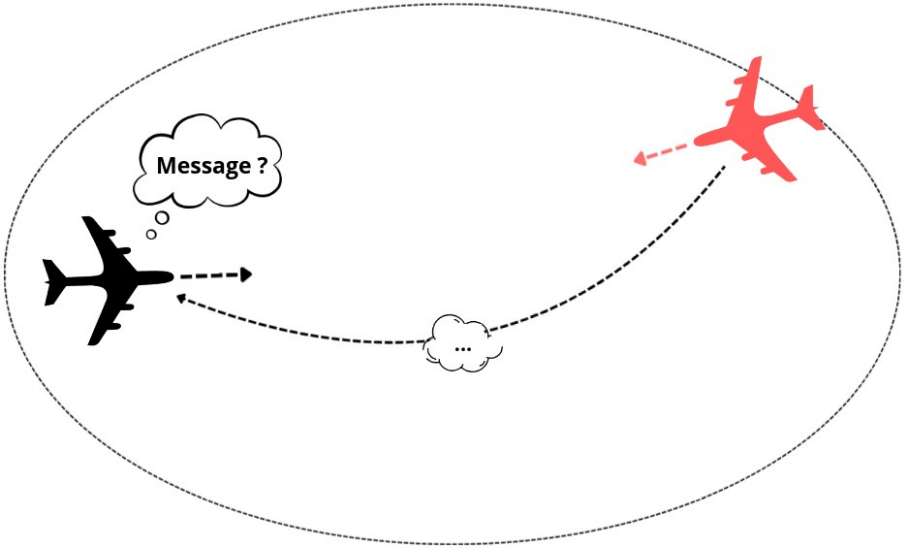
	01	02	03
Instantaneous Action	WR	SR	COC

→ **Final Action = SR**

The Action Selection : Cooperative-flight

The aircrafts communicate with each other and exchange the actions taken.

Before executing the chosen action, the system first checks if a message has been received :

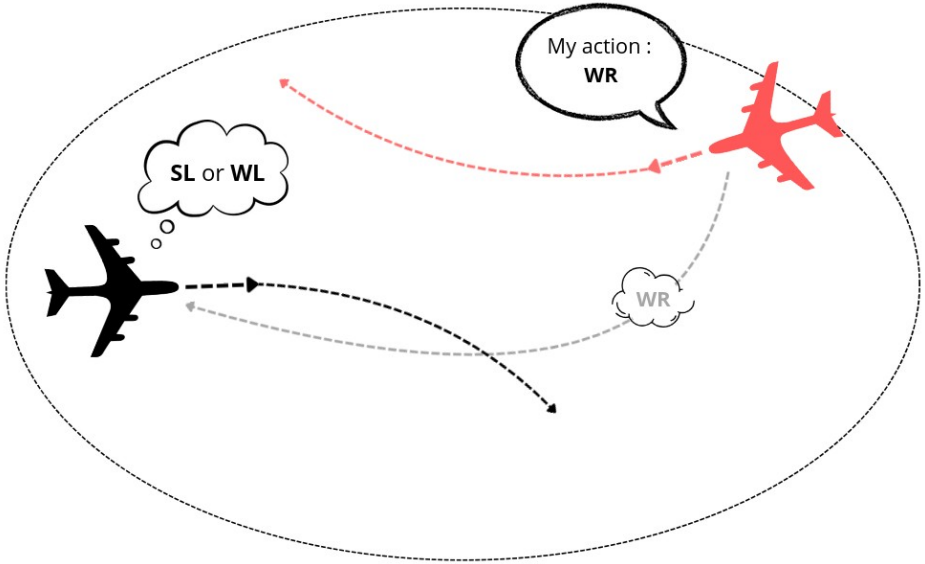


The Action Selection : Cooperative-flight

The aircrafts communicate with each other and exchange the actions taken.

Before executing the chosen action, the system first checks if a message has been received :

- Yes : a new complementary action to the one received must be chosen.

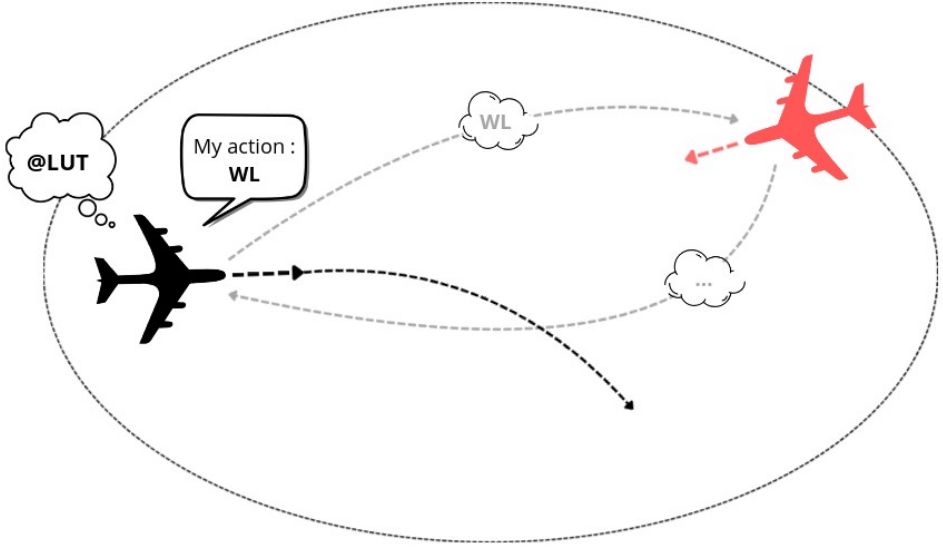


The Action Selection : Cooperative-flight

The aircrafts communicate with each other and exchange the actions taken.

Before executing the chosen action, the system first checks if a message has been received :

- No : the selected action is executed.



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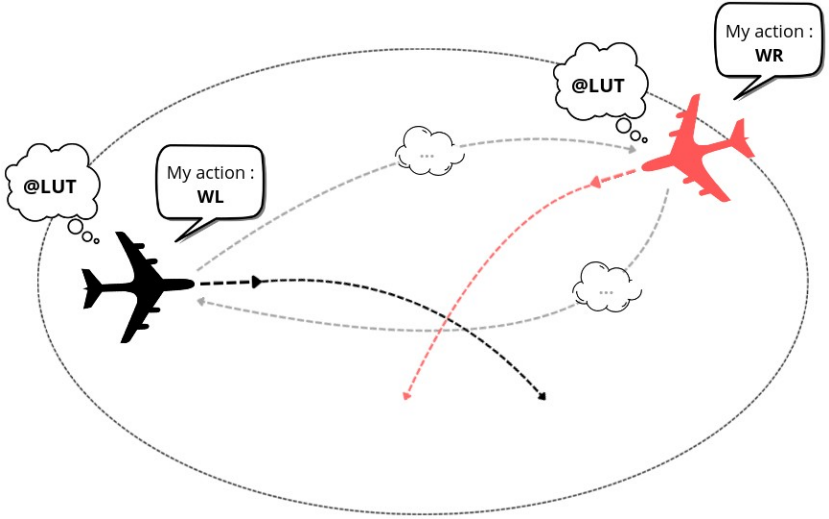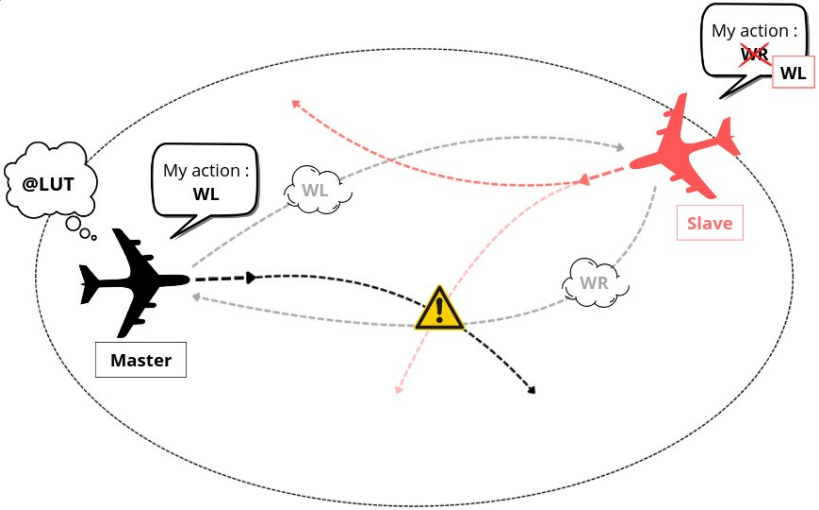
- If both make a decision before receiving any message, they botch proceed with their chosen action. BUT if these actions are incompatible, one must change.